CUDA-X: Unsupervised Domain-Adaptive Vehicle-to-Everything Collaboration via Knowledge Transfer and Alignment

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***Abstract*— Recently emerged vehicle-to-everything (V2X) perception has revealed great potential to overcome the limitation of single-vehicle intelligence aided by vigorous interaction among on-road agents, while prior endeavors are practically developed on parameter-specific simulation or configuration-dynamic real-world setting, overlooking the transferability across various scenarios. In this article, we propose unsupervised domain-adaptive vehicle-to-everything collaboration framework dubbed CUDA-X, which is built on top of a de facto collective model with key-point information exchange and instance adaptation. Specifically, collaborative knowledge transfer (CKT) is responsible for domain-agnostic feature reconstruction from nearby car or infrastructure by spatial–channel pooling operation in an elementwise manner. To promote the candidate alignment, a brand-new bin-based location correction (BLC) provides an auxiliary supervision for cross-dataset box refinement via residual coordinate encoding (RCE), and category-aware pooling alignment (CPA) is further designed for pulling the category-specific instance closer between source and target samples. We benchmark CUDA-X against the counterparts on four prevalent cooperative perception datasets, i.e., OPV2V, V2X-Sim, V2V4Real, and DAIR-V2X: it establishes the new state-of-the-art vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) performances regardless of simulation or reality. We expect that this appealing attempt would provide an in-depth insight into domain generalization in the context of multiagent perception, and the code is publicly available soon.**

***Index Terms*— 3-D object detection, autonomous driving, unsupervised domain adaptation (UDA), vehicle-to-everything (V2X) cooperative perception.**

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I. Introduction

Perception in the surroundings is pivotal for robust prediction and reasonable planning in the safety-critical system, i.e., autonomous driving (AV), contributing to a dramatic reduction of traffic conflicts and collisions. Recent developments in deep learning [1] facilitate the progress of individual perception, and remarkable achievements are observed in a large number of tasks including object detection [2], [3], [4], tracking [5], [6], [7], and segmentation [8], [9], [10]. Despite its advancements, single-vehicle perception is vulnerable to handle several practical challenges, e.g., occlusion, blind spot, over-the-horizon, and so on, resulting in catastrophic accidents or potential hazards.

Vehicle-to-everything (V2X) perception [11] opens a new door to solve the bottlenecks of individual intelligence, by means of data transmission and information sharing among agents. Generally, it is formulated as LiDAR-based 3-D cooperative object detection, and they could be roughly categorized into three folds according to different collaborative strategies. On the one hand, early fusion [12] aims to aggregate raw point cloud from nearby agents for data enhancement, while late collaboration [13] is responsible for high-quality candidate integration via postprocessing operation. Nonetheless, the former struggles to prohibit communication bandwidth and computational overhead during transmission, and the latter suffers from accumulated error and localization offset, thus neither of them not being preferred. On the other hand, intermediate fusion [14] is an appealing paradigm due to its better trade-off between accuracy and efficiency, which concentrates on informative region-of-interests (RoIs) aggregation for performance improvement. By leveraging the simulation (i.e., V2X-Sim [15] and OPV2V [16]) and real-world (e.g., V2V4Real [17] and DAIR-V2X [18]) datasets, a large collection of collaborations [19], [20], [21], [22], [23] is flourished to endow connected automated vehicles (CAVs) with broader sensing scope and longer probing range, demonstrating the strong robustness against challenging variations.

Nevertheless, domain gap is the long-standing issue for a safety-critical system when deployed on an unseen or

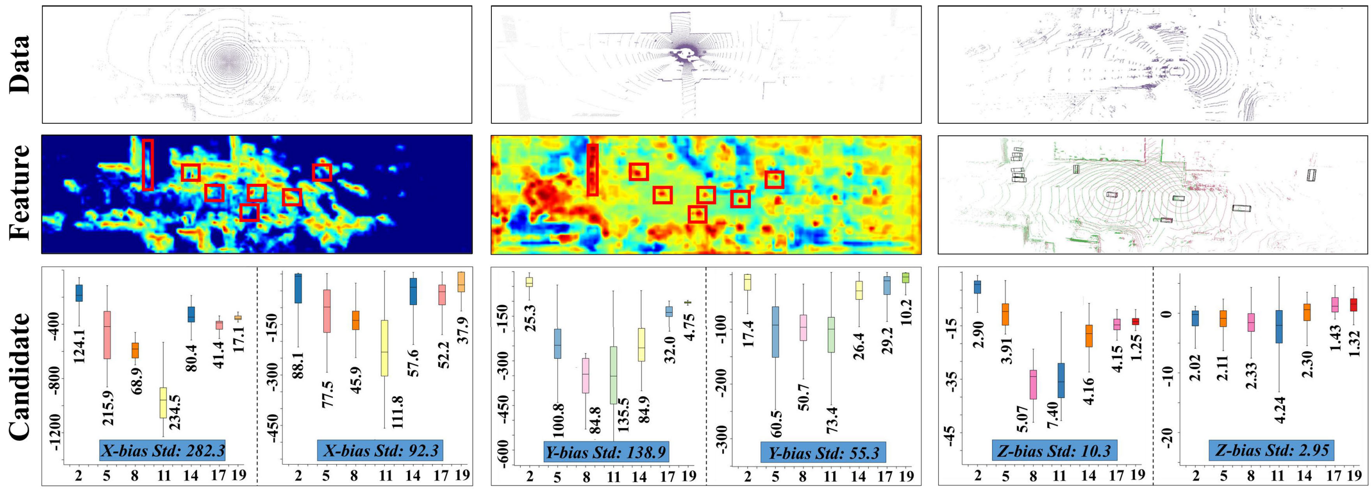

Fig. 1. Domain-discrepancy formulation in three perspectives. We denote $r1/c1$ as the first row/column for convenience: *1) data*—the sample of V2X-Sim ($r1c1$), OPV2V ($r1c2$), and V2V4Real ($r1c3$) is listed from the left to right column, and remarkable pattern differences could be evidently observed in sim2sim and sim2real conditions. Taking CoBEVT in sim2real (i.e., *OPV2V→V2V4Real*) as an example: *2) feature*—the attention maps after collaboration of *baseline* ($r2c1$) and *Oracle* ($r2c2$) are visualized on V2V4Real test set, as well as the corresponding GT ($r2c3$) respectively; compared to the potential targets with red box spotted by *Oracle*, it is nontrivial that *baseline* would introduce severe feature ambiguity and semantic confusion due to the cross-dataset shift; and *3) candidate*—boxplot of the residual between prediction (left *baseline*/right *Oracle*) with GT is depicted with respect to X-, Y-, and Z-parameters. During all training batches, we notice that box center predicted by the *baseline* is fluctuated with a larger standard deviation, e.g., 282.3 versus 92.3 in X, implying its poor and unstable regression when adapting to various samples.

unknown environment. Essentially, we conduct a pilot study on the impact of the domain discrepancy due to environmental variability and sensor differences on detection performance under vehicle-to-vehicle (V2V) perception, and formulate a new perspective of **Data** → **Feature** → **Candidate** to disclose the immense discrepancy between simulation to simulation (sim2sim) and simulation to reality (sim2real). As depicted in Fig. 1, data gaps between various datasets can be apparently observed in sensor type, scene layout, and beam pattern. Taking CoBEVT [21] in OPV2V→V2V4Real scenario for in-depth analysis, attention maps after collaboration of the *baseline* (no adaptation) and *Oracle* are visualized with the corresponding ground truth (GT) from the left to right column, respectively. Due to domain shift, it is nontrivial that the baseline would introduce serious feature confusion and semantic ambiguity in comparison to the possible targets with red box spotted by *Oracle*, resulting in enormous false positives. More importantly, boxplot of the X–Y–Z residual between the candidate (left *baseline*/right *Oracle*) with GT is calculated throughout each batches. It is concluded that candidate center predicted by the *baseline* is fluctuated dramatically with a larger standard deviation, e.g., 282.3 versus 92.3 in X, suggesting its unstable and unsatisfactory localization when generalizing to uncertain samples. It is consequently imperative to encourage a domain-adaptive pipeline of multiagent perception for mitigating performance degradation across varying conditions.

To these ends, we propose a novel unsupervised domain-adaptive Vehicle-to-everything Collaboration paradigm CUDA-X, from intermediate feature transfer to candidate instance alignment. Based on an arbitrary cooperative perception, collaborative knowledge transfer (CKT) is first introduced to exploit location-specific and semantic-aware cues for domain-invariant representation learning among nearby agents. The foreground embedding is split and highlighted from the fused map by a concurrent spatial–channel pooling architecture, and then we incorporate it with ego feature via an indexwise matrix multiplication. The discriminator is followed with a gradient reverse layer (GRL) [24] for information propagation and source/target sample recognition. To promote instance-level adaptation, bin-based location correction (BLC) is further designed in parallel with prediction header, which is responsible for box coordinate refinement and cross-dataset localization supervision. Due to the intractable optimization for continued space, residual coordinate encoding (RCE) is developed for dividing box coordinate into a group of even-interval bins and assigning a unique class according to its fallen position. With the help of bin-based discretization, we progressively update BLC prediction and meanwhile enforce an auxiliary supervision for distinguishing the candidate belonging to source or target by a discriminator. Moreover, category-aware pooling alignment (CPA) branch is constructed to condense the global significance of target via simple global average pooling (GAP) and multilayer perceptron (MLP), aiming for cross-domain category-specific instance alignment via a classifier.

Extensive experiments are conducted among V2X-Sim, OPV2V, V2V4Real, and DAIR-V2X benchmarks, and our proposed CUDA-X achieves the state-of-the-art V2V and vehicle-to-infrastructure (V2I) performance with respect to sim2sim, sim2real, and real2real adaptations, demonstrating its advancement and superiority. To our best knowledge, it is a thorough exploration of multiagent adaptive perception, and we expect that this work would propel the field of networked autonomous driving toward practicality and generalization.

II. Related Works

This section aims to review the related studies on LiDAR-based, cooperative, and domain-adaptive 3-D object detection.

A. LiDAR-Based 3-D Object Detection

LiDAR-based 3-D detection has received increasing attentions due to its valuable geometry structure information, and previous works can be mainly divided into point-based [25], [26], voxel-based [27], [28], [29], and point-voxel [30], [31] folds. Point-based detector (i.e., PointRCNN [32]) utilizes raw LiDAR data for pointwise feature learning via PointNet [33] or PointNet++ [34] backbone, and it produces classification score and seven-freedom box regression via a detection header. To handle the irregular and unordered data format, VoxelNet [35] divides the point cloud into equally spaced 3-D voxels and transforms a group of points within each voxel into a unified voxel-based feature to be processed by 3-D CNN efficiently. To leverage the advantage of both point and voxel representations, PV-RCNN [36] abstracts a small set of key points from 3-D voxel CNN, and RoI grid points are pooled into proposal-specific feature for object confidence and location estimation. In this work, we formulate cooperative perception as a voxel-based 3-D detection task for fast feature encoding and inference efficiency.

B. Cooperative 3-D Object Detection

Cooperative perception provides great potential to revolutionize the autonomous-driving paradigm, which aggregates complementary information across nearby agents to enhance the overall perception performance. Current studies are appropriately categorized into early data [12], intermediate feature [14], [37], and late candidate [13] collaboration. Specifically, early fusion suffers from cumbersome raw point processing, while error-accumulated issue in later pipeline might damage the final performance to some extent. Contrarily, intermediate fusion [20], [21], [22] is more favored due to its better trade-off between accuracy and communication bandwidth. Xu et al. [16] construct the first V2V perception benchmark OPV2V based on open-sourced CARLA [38] and SUMO [39] simulators, and design an attentive fusion method for multiagent object detection. DiscoNet [19] trains a teacher–student network for the shared feature aggregation, and simultaneously synthesizes a large-scale multiagent dataset V2X-Sim [15]. To accelerate the practical development of V2V perception, V2V4Real [17] is collected as the first real-world multimodal dataset for such cooperative 3-D detection, tracking, and domain adaptation tasks. V2X-ViT [40] introduces the heterogeneous multiagent graph via V2X communication for spatial relationship among different agent types. DAIR-V2X [18] provides real-world vehicle-infrastructure samples for stimulating heterogeneous cooperative perception. In this article, we follow these well-designed collaborations, and aim to propose domain-adaptive modules to bridge the distributional gap.

C. Domain-Adaptive 3-D Object Detection

Unsupervised domain adaptation (UDA) [41] reveals a promising solution to generalize a well-trained model to the unseen scenario while circumventing laborious labeling process. Prior efforts either adopt self-training strategy [42], [43] to generate pseudo labels for unannotated target data, or mitigate the domain divergence in an adversarial learning manner [44], [45]. ST3D [42] conducts pseudo-label updating with quality-aware triplet memory bank and develops curriculum data augmentation to avoid overfitting to easy sample in newly annotated data. Conversely, SPG [46] generates object-level semantic points for enhancing the reliability of LiDAR 3-D detector against distributional gaps. To bridge the gap between the simulated data and real-world data, Xiao et al. [47] leverage a novel point cloud translator to close the synthetic LiDAR to true point data, while DODA [48] is a data-oriented domain adaptation framework to pattern and context differences across domains. Despite its exhaustive studies and significant progresses, domain adaptation in multiagent 3-D detection remains an open problem. Concurrently, S2R-ViT [49] launches the first V2V adaptive perception that incorporates an uncertainty-aware transformer with feature adapter for data-irrelevant generalization ability, while it restricts with the sim2real setting among vehicles. In this work, we delve into a thorough multiagent perception in consideration of sim/real and vehicle/roadside adaptations.

III. Methodology

In this article, we investigate a novel unsupervised domain-adaptive Vehicle-to-everything collaborative perception framework named as CUDA-X. Herein, problem statement is formulated at first, and we technically introduce the architectural overview with crucial ingredients, including CKT, BLC, and CPA.

A. Problem Formulation

In the context of UDA, an annotated source dataset is presented as $D^s = \{(X_i^s, Y_i^s)|_{i=1}^{n_s}\}$, where $X_i^s = (x_i, y_i, z_i, r_i)$ denotes the location parameters along the x-, y-, and z-axes and reflectance intensity r for each data X_i^s, Y_i^s implies the label information, and n_s is the total number of point clouds. Likely, the unlabeled target domain is marked as $D^t = \{(X_j^t)|_{j=1}^{n_t}\}$ with n_t examples. Consequently, our goal is to obtain a meticulous-performance multiagent cooperative detector that could generalize to the unseen samples well with marginal accuracy decline.

B. Overview of CUDA-X

Fig. 2 demonstrates the architectural overview of CUDA-X: it encompasses a mainstream cooperative perception framework and three domain-adaptive branches. Noted that we follow the pipeline of intermediate collaboration with essential *feature extraction*, *compression and sharing*, *information fusion* and *prediction header* sections.

After pose broadcasting and exchange among agents, LiDAR backbone is utilized for point *feature extraction*: given

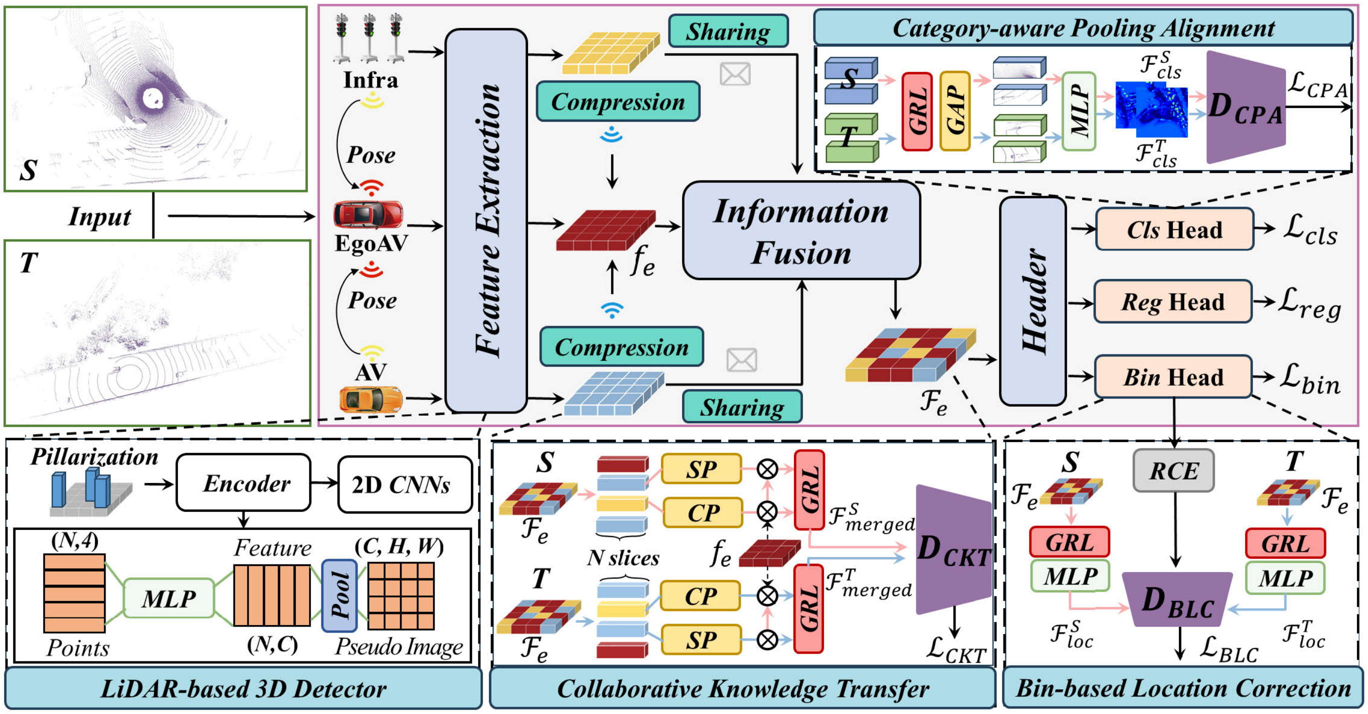

Fig. 2. Architectural overview of our proposed CUDA-X. In the mainstream framework, an intermediate collaborative perception paradigm typically experiences *feature extraction* with LiDAR-based 3-D detector, *compression and sharing*, *information fusion*, and *prediction header*. For domain-adaptive task, *1) CKT*—multiagent fused representation $\mathcal{F}_e$ is incorporated with the ego map f_e after slicing and spatial (SP)–channel (CP) pooling operations, and a discriminator $\mathcal{D}_{\text{CKT}}$ is adopted for source/target sample recognition; *2) BLC*—RCE is proposed for label discretization, which is referred to as an auxiliary component for both box refinement and cross-domain localization supervision via $\mathcal{D}_{\text{BLC}}$; *3) CPA*—it is responsible for fine-grained instance-level feature distillation via GAP and MLP, promoting for category-specific candidate alignment by a classifier $\mathcal{D}_{\text{CPA}}$. Noted that different colors demonstrate the gradient flow of source/target sets via a GRL, and the overall objectives consist of detection loss ($\mathcal{L}_{\text{cls}}$, $\mathcal{L}_{\text{reg}}$, and $\mathcal{L}_{\text{bin}}$) and domain-adaptation loss ($\mathcal{L}_{\text{CKT}}$, $\mathcal{L}_{\text{BLC}}$, and $\mathcal{L}_{\text{CPA}}$). Best viewed in color.

M collaborators denoted as $C = \{(c_m)|_{m=1}^{M}\}$, it produces a group of maps $\mathcal{F} = \{(f_m)|_{m=1}^{M}\}$. Noted that an off-the-shelf *LiDAR-based 3-D Detector* (i.e., PointPillars [28]) could be viewed as feature encoder. Subsequently, *compression and sharing* with each other is performed by projecting these intermediate maps into a unified reference coordinate via an encoder–decoder architecture, and we further aggregate them reasonably through an *information fusion* module, forming an ego-centric map $\mathcal{F}_e$. Finally, *prediction header* outputs classification score s_p and box regression $(x_p, y_p, z_p, w_p, h_p, l_p, \theta_p)$ for each object, where (x_p, y_p, z_p) denotes the center coordinate prediction, (w_p, h_p, l_p) implies object size in 3-D space, and θ_p is the yaw angle, respectively.

On top of it, three adaptive modules are customized for alleviating cross-domain discrepancy issue: CKT is responsible for domain-agnostic representation learning from both spatial and channel dimensions; BLC is designed for both box refinement and cross-domain localization rectification; meanwhile, CPA is further introduced to exploit the category-specific information significance for instance-level candidate alignment.

C. Collaborative Knowledge Transfer

Due to multifacet gaps in sensor type, reflectance pattern, road layout, and so on, it is struggled for domain-agnostic feature learning, and performance drop would be unavoidably occurred when transferring a well-trained model to the unseen target domain. Hence, domain-invariance collaborative adaptation remains to be discussed. A naïve solution is to directly adapt the fused result from one to another, but it fails to capture geometrical and semantic cues which are crucial for domain discrimination.

In this work, we propose CKT module as depicted in Fig. 3, a split-merge architecture that exploits spatial–channel information exhaustively across CAVs from distinct perspectives for ego-centric map enhancement. Given the fused and ego representations $(\mathcal{F}_e, f_e) \in \mathbb{R}^{C\times H\times W}$ where (C, H, W) denotes the feature channel and spatial size, we split them uniformly into N slices $(\mathcal{F}^n_{\text{es}}, f^n_{\text{es}}) \in \mathbb{R}^{(C/N)\times H\times W}$ along the channel dimension. Taking $(\mathcal{F}^1_{\text{es}}, f^1_{\text{es}})$ as example, $\mathcal{F}^1_{\text{es}}$ experiences spatial adaptive pooling with 1×1 convolution and linear channel pooling operations concurrently, producing geometry-aware and semantics-aware vectors $(\mathcal{F}^{(1,s)}_{\text{es}}, \mathcal{F}^{(1,c)}_{\text{es}})$ respectively. The ego slice f^1_{es} is further multiplied sequentially with spatial and semantic embeddings according to the corresponding index, and we obtain a reconstructed sequence after channel shuffle with a dimension of $(C/N) \times H \times W$. In this way, N slices are repeatedly updated with great importance of geometry and semantic cues, and we merge them in a channelwise concatenation

$$\mathcal{F}^{(n,s)}_{\text{es}} = SP_{1\times 1}(\mathcal{F}_{\text{es}}) \in \mathbb{R}^{\frac{C}{N}\times 1\times 1} \tag{1}$$

$$\mathcal{F}^{(n,c)}_{\text{es}} = CP_{\text{linear}}(\mathcal{F}_{\text{es}}) \in \mathbb{R}^{1\times H\times W} \tag{2}$$

$$\mathcal{F}^{n}_{\text{rec}} = \text{Shuffle}\Big(f^n_{\text{es}} \bigotimes \mathcal{F}^{(n,s)}_{\text{es}} \bigotimes \mathcal{F}^{(n,c)}_{\text{es}}\Big) \tag{3}$$

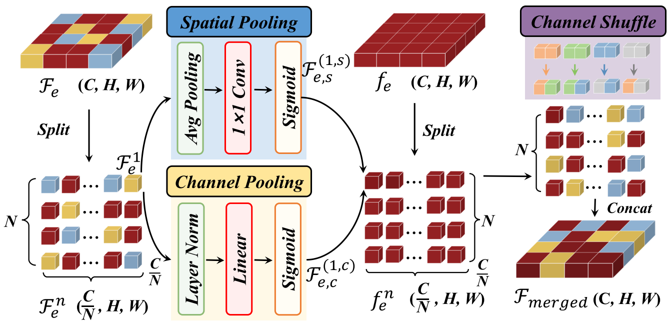

Fig. 3. Schematic of knowledge collaboration in CKT. The multiagent fused representation $\mathcal{F}_e$ is split into N slice along the channel, producing $\mathcal{F}_e^n$ with a dimension of $(C/N) \times H \times W$. We perform spatial–channel pooling operations separately, and merge them into the sliced ego map f_e according to feature index. After channel shuffle, we generate the reconstructed vector $\mathcal{F}_{\text{merged}}$ with a integration of both cross-agent and intra-agent location correlation and semantics. Best viewed in color.

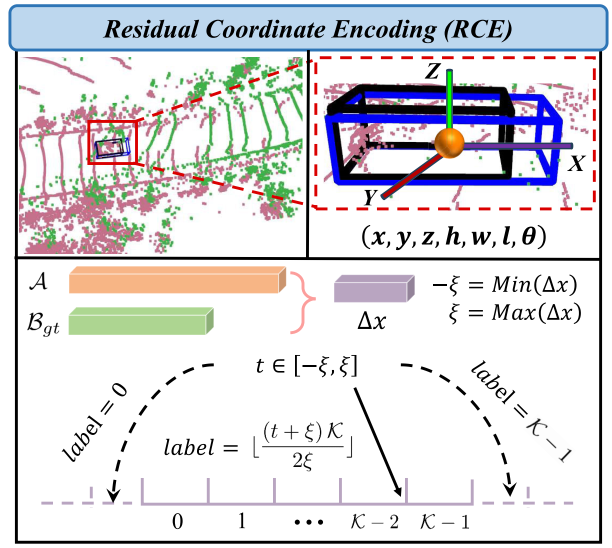

Fig. 4. Schematic of RCE in BLC. Given seven-parameter GT $\mathcal{B}$ and anchor $\mathcal{A}$ boxes, residual Δ is calculated with respect to each coordinate, and the lower $-\xi$ and upper ξ bounds are determined by its minimum and maximum. In the case of $x = t$, we equally divide $[-\xi, \xi]$ into $\mathcal{K}$ intervals, and a unique label is generated according to the fallen position: if $-\xi < t < \xi$, it is tagged as $\lfloor((t + \xi)\mathcal{K}/2\xi)\rfloor$; otherwise, we assign 0 or $\mathcal{K} - 1$ when it drops outside the range. Best viewed in color.

$$\mathcal{F}_{\text{merged}} = \mathcal{F}_{\text{rec}}^1 \bigoplus \mathcal{F}_{\text{rec}}^2 \cdots \bigoplus \mathcal{F}_{\text{rec}}^n, \quad n = 1, \ldots, N \tag{4}$$

where $SP_{1\times1}(\cdot)$, $CP_{\text{linear}}(\cdot)$, and Shuffle$(\cdot)$ stand for spatial/channel pooling calculation and shuffle transformation, and $\mathcal{F}_{\text{rec}}^n$ denotes the reconstructed vector, respectively. Due to the masking effect of spatial–channel pooling mechanism, the ultimately merged map $\mathcal{F}_{\text{merged}}$ is integrated location-aware correlation across vehicles with intra-agent semantics, thus being favorable for domain-adaptive alignment and multiagent perception.

The domain discriminator $\mathcal{D}_{\text{CKT}}$ is responsible for distinguishing the sample belonging to the source or target dataset, and GRL is implemented for optimizing the max–min problem jointly in an adversarial manner. Generally, CKT loss $\mathcal{L}_{\text{CKT}}$ is formulated as binary cross-entropy (BCE) function: M indicates the whole cooperative perception framework, and $\mathcal{F}_{\text{merged}}^{(S,T)}$ denotes the merged feature from source or target domain

$$\max_{M} \min_{\mathcal{D}_{\text{CKT}}} \mathcal{L}_{\text{CKT}} = \mathbb{E}\Big[\mathcal{D}_{\text{CKT}}\Big(\mathcal{F}_{\text{merged}}^S\Big)\Big] + \mathbb{E}\Big[1 - \mathcal{D}_{\text{CKT}}\Big(\mathcal{F}_{\text{merged}}^T\Big)\Big]. \tag{5}$$

D. Bin-Based Location Correction

During model prediction, box coordinate is sensitive to scenario variance when adapting to the unknown world. An alternative intuition is to conduct location improvement from regression branch, while continued prediction value is hard to be rationally optimized. To this end, we approach a brand-new BLC header in parallel with the original architecture, which is regarded as an auxiliary component for box coordinate rectification in mainstream detection framework and cross-dataset localization supervision in domain-adaptive branch simultaneously.

1) Bin-Based Label Discretization: In essence, coordinate refinement is formulated as a multilabel classification problem and our primary task is to divide continuous space into discrete boundary with binwise label. Considering seven-degrees-of-freedom (7-DoF) offset continuous prediction in the vanilla regression, a novel RCE method is proposed for label discretization. Motivated by PointPillars, residuals Δ between each GT $\mathcal{B}_{\text{gt}}$ and the anchor box $\mathcal{A}$ are first calculated for each variable, the minimum and maximum of which are considered as the lower and upper bounds of the corresponding coordinate. As illustrated in Fig. 4, we further divide the range into $\mathcal{K}$ folds, and produce a particularly discretized label for each GT coordinate by observing in which piece it falls. For instance, assuming the residual $\Delta x \in [-\xi, \xi]$, there are $\mathcal{K}$ intervals with a length of $(2\xi/\mathcal{K})$. Given a GT box located at $x = t$, the whole process can be mathematically described as follows:

$$\text{RCE} = \begin{cases} 0, & -\xi <= t \\ \left\lfloor \dfrac{(t+\xi)\mathcal{K}}{2\xi} \right\rfloor, & -\xi < t < \xi \\ \mathcal{K} - 1, & t >= \xi \end{cases} \tag{6}$$

where $\lfloor \cdot \rfloor$ denotes the floor function. If x is out of the range (i.e., $t <= -\xi$ or $t >= \xi$), we tag it with label $\mathbf{0}$ or $\boldsymbol{\mathcal{K} - 1}$.

In mainstream branch, BLC header predicts separate 3-D-box parameters $\mathcal{B}_p \in \mathbb{R}^{\mathcal{K}\times 7}$ via three feed forward network (FFN) layers, conditioned on multiagent collaborative feature $\mathcal{F}_e \in \mathbb{R}^{C\times H\times W}$. During model training, the objective of BLC is to make the predictor approach GT label, and we adopt cross-entropy (CE) function as bin-based detection loss $\mathcal{L}_{\text{bin}}$ for supervision. By this way, the feature-to-location mapping is constructed between the source and target domains.

For domain-adaptive module, ego-centric fused map $\mathcal{F}_e \in \mathbb{R}^{C\times H\times W}$ undergoes a GRL and the same FFN design, and a discriminator $\mathcal{D}_{\text{BLC}}$ judges the source/target prediction $\mathcal{F}_{\text{loc}}^{(S,T)}$ with the help of CE-based adversarial loss $\mathcal{L}_{\text{BLC}}$

$$\max_{M} \min_{\mathcal{D}_{\text{BLC}}} \mathcal{L}_{\text{BLC}} = \mathbb{E}\big[\mathcal{D}_{\text{BLC}}\big(\mathcal{F}_{\text{loc}}^S\big)\big] + \mathbb{E}\big[1 - \mathcal{D}_{\text{BLC}}\big(\mathcal{F}_{\text{loc}}^T\big)\big]. \tag{7}$$

E. Category-Aware Pooling Alignment

Complex and dynamic environment typically involves a diversity of noises, i.e., background interference, semantic ambiguity, pattern confusion, and so on, causing semantic negative transfer and unsatisfactory recognition performance. In this article, we further develop a CPA branch for fine-grained instance-level feature distillation and adaptation.

As presented in Fig. 2, classification header with three FFNs is in charge of category score prediction from the joint ego map $\mathcal{F}_e \in \mathbb{R}^{C\times H\times W}$, forming a potential instance map $\mathcal{F}_s$ sized $H \times W \times C$. After a GRL, global averaging pooling (GAP($\cdot$)) is enforced to halve the feature scale, and we squeeze it into a 2-D RoI sequence $\mathcal{F}_s \in \mathbb{R}^{H\times(W\times C/2\times 2)}$ along the W-axis. Finally, MLP (MLP($\cdot$)) is utilized to capture the global response $\mathcal{F}_{\text{cls}}$ with great importance of category-agnostic instance information. The whole processing is formally concluded as follows:

$$\mathcal{F}_{\text{cls}} = \text{MLP}(\text{GAP}(\mathcal{F}_s)) \in \mathbb{R}^{H\times 1}. \tag{8}$$

Overall, CPA module underlines instance-level feature details by attending to foreground object more, which suggests sufficient robustness and adaptation across datasets. Followed by a discriminator $\mathcal{D}_{\text{CPA}}$, we could recognize the positive sample owned to source or target domain, and category-aware pooling adaption loss $\mathcal{L}_{\text{CPA}}$ is still based on CE formulation as follows:

$$\begin{aligned}\max_{M}\min_{\mathcal{D}_{\text{CPA}}} \mathcal{L}_{\text{CPA}} &= \mathbb{E}\big[\mathcal{D}_{\text{CPA}}\big(\mathcal{F}^{S}_{\text{cls}}\big)\big]\\ &\quad + \mathbb{E}\big[1 - \mathcal{D}_{\text{CPA}}\big(\mathcal{F}^{T}_{\text{cls}}\big)\big]\end{aligned} \tag{9}$$

where $\mathcal{F}^{(S,T)}_{\text{cls}}$ denotes the category prediction from source/target domain.

F. Training Loss

The training objectives of our framework totally contains detection loss $\mathcal{L}_{\text{det}}$ and domain-adaptive loss $\mathcal{L}_{\text{da}}$. On one hand, $\mathcal{L}_{\text{det}}$ includes classification loss $\mathcal{L}_{\text{cls}}$ with focal loss, box regression loss $\mathcal{L}_{\text{reg}}$ with smooth-$L1$ loss, as well as bin-based loss $\mathcal{L}_{\text{bin}}$ with CE function. On the other hand, we formulate $\mathcal{L}_{\text{CKT}}$, $\mathcal{L}_{\text{BLC}}$, and $\mathcal{L}_{\text{CPA}}$ for domain-adaptive supervision

$$\begin{aligned}\mathcal{L} &= \mathcal{L}_{\text{det}} + \mathcal{L}_{\text{da}}\\ &= \big(\mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{reg}} + \lambda_1\mathcal{L}_{\text{bin}}\big)\\ &\quad + \lambda_2(\mathcal{L}_{\text{CKT}} + \mathcal{L}_{\text{BLC}} + \mathcal{L}_{\text{CPA}})\end{aligned} \tag{10}$$

where $\lambda_1 = \lambda_2 = 0.1$ denote the balance coefficients for training stability.

IV. Experiments

In this section, empirical study is performed to measure the domain adaptation across different benchmarks, i.e., V2X-Sim, OPV2V, V2V4Real, and DAIR-V2X, and ablation analysis is further conducted to verify the effectiveness and contributions of each component.

A. Cooperative Perception Datasets

1) OPV2V [16]: It is a large-scale V2V perception benchmark built on the top of CARLA and SUMO simulators, which includes over 70 diverse scenarios from eight towns, with 11 464 frames and 232 913 labeled boxes. Noted that each scene contains on average 2.89 CAVs, each of which is equipped with a 64-channel LiDAR sensor for $10HZ$ recorded frequency. During model training, the train/val/test splits are divided into 6464/1981/2719 samples, respectively.

2) V2X-Sim [15]: Similarly, V2X-Sim is purely constructed on CARLA *Town05* map with multilane and cross-junction conditions. It consists of 10 000 frames with a 32-channel LiDAR recorded at 0.2 s, and 2$\sim$5 collaboration agents are considered for each scene. As commonly done, we split 23 500/3100 examples for model training and testing.

3) V2V4Real [17]: It is the first real-world dataset for V2V perception that acquires 20 000 32-beam LiDAR frames with more than 200 000 box annotations for five categories, e.g., *Car*, *Bus*, *Van*, and so on. Therefore, object size in length, width, and height spreads broadly from 2.5 to 23 m, from 1.5 to 4.5 m, and from 1 to 4.5 m, respectively, demonstrating its diversity and abundance. We partition the whole dataset into training/validation/test set with 14 210/2000/3986 frames.

4) DAIR-V2X [18]: As the pioneered V2I perception benchmark encompassed 28 selective intersections in Beijing High-level Autonomous Driving Demonstration Area, DAIR-V2X comprises 9311 pairs with collective annotations, each of which is obtained from both vehicle–roadside LiDAR sensors. The overall frames are stratified into train/val/test groups with a proportion of 5:2:3.

B. Implementation Details

We provide a comprehensive investigation of V2X domain generalization in both simulation and reality: 1) V2V Adaptation—sim2sim: V2X-Sim $\rightarrow$ OPV2V and sim2real: OPV2V $\rightarrow$ V2V4Real and 2) V2I Adaptation—sim2real: OPV2V $\rightarrow$ DAIR-V2X and real2real: V2V4Real $\rightarrow$ DAIR-V2X.

1) V2X-Sim $\rightarrow$ OPV2V: The proposed method is trained on both labeled samples of V2X-Sim and OPV2V, and adaptation performance is evaluated on OPV2V test set. Therefore, all data are transformed into V2X-Sim format, and we utilize PointPillars [28] as feature extractor, with the point range of $[-32, 32]$ m, $[-32, 32]$ m, and $[-8, -3]$ m along the x-, y-, and z-axes, respectively. The model is totally trained with Adam [50] optimizer of 1e^{-4} weight decay for 50 epochs, and multistep learning rate strategy is introduced by an initial 0.001 with $\gamma = 0.5$ at [50, 100, 150, 200] steps. The category and NMS IoU thresholds are 0.4 and 0.1.

2) OPV2V $\rightarrow$ V2V4Real: Our method also evaluates the adaptation performance from OPV2V simulation to the real-world V2V4Real under the same data style. Similarly, PointPillars is adopted as point extractor within the range of $[-140, 140]$ m along the x-axis, $[-40, 40]$ m along the y-axis, and $[-5, 3]$ m along the z-axis. The model is optimized by Adam and cosine-anneal warm strategy [51] for 45 epochs, with an initial learning rate of 0.001 and 1e^{-4}

TABLE I

COMPARISONS WITH SOTAS IN CASE OF **V2X-SIM → OPV2V (SIM2SIM V2V)**. NOTED THAT *Oracle* IN RED FONT SUGGESTS THE MODEL IS TRAINED AND TESTED ON THE SAME TARGET DOMAIN, AND WE DRAW THE RESULTS ACHIEVED BY OUR PROPOSED METHOD WITH BLUE-BOLD COLOR. THE PERFORMANCE GAINS (+)/DROPS (−) ARE LISTED IN THE BRACKET

V2XSim → OPV2V (sim2sim V2V)	When2comm		V2VNet		DiscoNet	
	AP@50 (%)	AP@70 (%)	AP@50 (%)	AP@70 (%)	AP@50 (%)	AP@70 (%)
w./o	29.1	8.7	39.1	8.7	47.7	13.2
w./GRL	30.1 (+1.0)	7.7 (−1.0)	39.5 (+0.4)	9.6 (+0.9)	48.4 (+0.7)	12.6 (−0.6)
w./AdvGRL	32.0 (+2.9)	7.6 (−1.1)	41.2 (+2.1)	9.6 (+0.9)	50.4 (+2.7)	14.1 (+0.9)
***w.*/Ours**	**42.7 (+13.6)**	**9.7 (+1.0)**	**48.6 (+9.5)**	**10.4 (+1.7)**	**51.4 (+3.7)**	**14.6 (+1.4)**
Oracle	55.1 (+26.0)	23.3 (+14.6)	59.0 (+19.9)	24.7 (+16.0)	55.5 (+7.8)	25.4 (+12.2)

weight decay. The positive/negative/score/IoU thresholds are 0.6/0.45/0.2/0.15, respectively.

3) OPV2V/V2V4Real → DAIR-V2X: To offer an in-depth insight on heterogeneous-agent versatility, the method is further generalized from V2V to V2I collaborations. Unless otherwise stated, it is progressively updated with cosine-anneal learning strategy for 35 epochs, and other technical details follow the **OPV2V→V2V4Real** settings for simplicity, e.g., perception range, IoU threshold, and so on. All experiments are implemented on a single NVIDIA RTX3090 GPU. In CKT module, the number of spatial convolution and channel linearity are 8 and 1; for BLC branch, three FFNs include (256, 256, 256) channels, and bin-based header outputs a $5 \times 6 \times 2$ vector, where $\mathcal{K} = 5$, $6 = (x, y, z, w, h, l)$ denotes the center coordinate and 3-D box size, and two implies two anchors with 0° or 90° orientations; CPA contains three MLP layers with a dimension of (88, 1024, 1024). We adopt When2com [11], V2VNet [14], DiscoNet [19], F-Cooper [37], AttFuse [16], V2X-ViT [40], and CoBEVT [21] as collaboration frameworks, and append the baseline GRL [17], adversarial GRL (AdvGRL) [52], S2R-AFA [49], and DUSA [53], our proposed method and *Oracle* (train and test on the same dataset) for comparisons. Detailed implementations can refer to *OpenCOOD*[1] and *Coperception*[2] benchmarks.

C. Comparisons With SOTAs

1) V2X-Sim → OPV2V: As tabulated in Table I, obvious performance degradation can be observed with respect to all cooperative preceptors when transferring from V2X-Sim to OPV2V set, i.e., 26.0% AP@50 and 14.6% AP@70 drops for When2com [11], 19.9% AP@50 and 16.0% AP@70 reductions in V2VNet [14], and so on.

Previous works achieve a marginal accuracy improvement: at 0.5 IoU threshold, GRL [17]/AdvGRL [52] methods merely offer 1.0%/2.9%, 0.4%/2.1%, and 0.7%/2.7% AP boosts for When2com, V2VNet, and DiscoNet [19], respectively; it is however an accuracy fluctuation in 0.7 IoU level, which implies the limitations in handling domain shift issue. On the contrary, CUDA-X shows a great domain generalization and consistently outperforms other counterparts over a remarkable margin. Particularly, it provides 13.6%/9.5%/3.7%**AP**@50 gains for three collaborators without adaptation, and further promotes 1.0%∼1.7%**AP**@70, demonstrating its advancement and universality.

2) OPV2V → V2V4Real: The sim2real V2V adaption result is presented in Table II. Comparably, the proposed CUDA-X still enhances the adaptation performance of multiagent baselines across datasets, which offers 11.1%∼21.7%**AP**@50 and 3.4%∼6.9%**AP**@70 increments with various collaborations. Compared to the SOTA alternatives, it steadily upgrades an accuracy of 2.4%∼8.3% and 1.2%∼4.8% at both 0.5/0.7 IoU levels against AdvGRL method, suggesting the superiority and versatility of our method. Notably, the state-of-the-art S2R-UViT [49] reports 44.1% AP@50 and 17% AP@70 results, which are on par with or even slightly superior to CUDA-X in sim2real V2V setting. We presume that the powerful uncertainty vision transformer (UViT) might make effect on domain-invariance representation learning, thereby contributing to the outstanding accuracy. We would delve into a fair comparison and in-depth analysis until its code releases.

3) OPV2V → DAIR-V2X: Surprisingly, our method exhibits profound performance when transferring V2V experience to V2I scene. As evidenced in Table III, it establishes a new state-of-the-art result of 35.5%/37.0%**AP**@50 and 17.7%/19.9%**AP**@70 at 0.5/0.7 thresholds in integration with CoBEVT/V2X-ViT, which further rivals two alternatives by 2.0%∼5.4% and 1.5%∼6.0% AP respectively. It is noteworthy that even compared with the latest state-of-the-art method DUSA [53], our method still yields a 0.5% and 1% improvement at 0.5/0.7 thresholds, respectively. Due to the transparent difficulty in varied scenarios (sim2real) and heterogeneous collaboration (V2V → V2I), remarkable improvements highlight the significant knowledge sharing and comprehensive understanding of V2X setting.

4) V2V4Real → DAIR-V2X: Table IV eventually testifies the heterogeneous-agent adaptation in the real world. Equipped with our CUDA-X, CoBEVT and V2X-ViT constantly narrow the domain gap with an AP growth of 9.6%/6.8% and 7.1%/4.7% compared to no-adaptation baseline, and meanwhile excel GRL/AdvGRL over an explicit advantage. The overwhelming outcome hints the applicable and practical generalization among these disparate agents.

[1]https://github.com/DerrickXuNu/OpenCOOD

[2]https://github.com/coperception/coperception

TABLE II

COMPARISONS WITH SOTAS IN CASE OF **OPV2V** → **V2V4REAL** (**SIM2REAL V2V**). NOTED THAT *Oracle* IN RED FONT SUGGESTS THE MODEL IS TRAINED AND TESTED ON THE SAME TARGET DOMAIN, AND WE DRAW THE RESULTS ACHIEVED BY OUR PROPOSED METHOD WITH BLUE-BOLD COLOR. THE PERFORMANCE GAINS (+)/DROPS (−) ARE LISTED IN THE BRACKET

OPV2V → V2VReal (sim2real V2V)		
Methods	*V2V4Real Test Set*	
	AP@50 (%)	*AP@70 (%)*
AttFuse *w.*/o	22.5	9.4
AttFuse *w.*/GRL	34.2 (+11.7)	10.9 (+1.5)
AttFuse *w.*/AdvGRL	35.9 (+13.4)	11.5 (+2.1)
AttFuse *w.*/Ours	**44.2 (+21.7)**	**16.3 (+6.9)**
AttFuse (*Oracle*)	64.5 (+42.0)	34.3 (+24.9)
F-Cooper *w.*/o	23.6	9.1
F-Cooper *w.*/GRL	29.6 (+6.0)	7.6 (−1.5)
F-Cooper *w.*/ AdvGRL	37.7 (+14.1)	9.3 (+0.2)
F-Cooper *w.*/Ours	**40.1 (+16.5)**	**12.5 (+3.4)**
F-Cooper (*Oracle*)	60.7 (+37.1)	31.8 (+22.7)
V2X-ViT *w.*/o	27.4	10.3
V2X-ViT *w.*/GRL	39.4 (+12.0)	14.8 (+4.5)
V2X-ViT *w.*/AdvGRL	42.1 (+14.7)	15.9 (+5.6)
V2X-ViT *w.*/Ours	**45.0 (+17.6)**	**17.1 (+6.8)**
V2X-ViT (*Oracle*)	64.9 (+37.5)	36.9 (+26.6)
CoBEVT *w.*/o	33.4	13.0
CoBEVT *w.*/GRL	39.8 (+6.4)	13.9 (+0.9)
CoBEVT *w.*/AdvGRL	41.5 (+8.1)	16.1 (+3.1)
CoBEVT *w.*/Ours	**44.5 (+11.1)**	**17.5 (+4.5)**
CoBEVT (*Oracle*)	66.5 (+33.1)	36.0 (+23.0)
S2R-UViT *w.*/o	36.7	13.8
S2R-UViT *w.*/GRL	41.4 (+4.7)	14.1 (+0.3)
S2R-UViT *w.*/AdvGRL	41.4 (+4.7)	15.7 (+1.9)
S2R-UViT *w.*/S2R-AFA	**44.1 (+7.4)**	**17.0 (+3.2)**

TABLE III

COMPARISONS WITH SOTAS IN CASE OF **OPV2V** → **DAIR-V2X** (**SIM2REAL V2I**). NOTED THAT *Oracle* IN RED FONT SUGGESTS THE MODEL IS TRAINED AND TESTED ON THE SAME TARGET DOMAIN, AND WE DRAW THE RESULTS ACHIEVED BY OUR PROPOSED METHOD WITH BLUE-BOLD COLOR. THE PERFORMANCE GAINS (+)/DROPS (−) ARE LISTED IN THE BRACKET

OPV2V → DAIR-V2X (sim2real V2I)		
Methods	*V2V4Real Test Set*	
	AP@50 (%)	*AP@70 (%)*
CoBEVT *w.*/o	19.8	13.0
CoBEVT *w.*/GRL	31.5 (+11.7)	14.6 (+1.6)
CoBEVT *w.*/AdvGRL	33.5 (+13.7)	16.2 (+3.2)
CoBEVT *w.*/Ours	**35.5 (+15.7)**	**17.7 (+4.7)**
CoBEVT (*Oracle*)	60.9 (+41.1)	43.8 (+30.8)
V2X-ViT *w.*/o	29.5	10.9
V2X-ViT *w.*/GRL	33.0 (+3.5)	13.9 (+3.0)
V2X-ViT *w.*/AdvGRL	31.6 (+2.1)	15.0 (+4.1)
V2X-ViT *w.*/DUSA	36.5 (+7.0)	18.9 (+8.0)
V2X-ViT *w.*/Ours	**37.0 (+7.5)**	**19.9 (+9.0)**
V2X-ViT (*Oracle*)	56.1 (+26.6)	40.9 (+30.0)

TABLE IV

COMPARISONS WITH SOTAS IN CASE OF **V2V4REAL** → **DAIR-V2X** (**REAL2REAL V2I**). NOTED THAT *Oracle* IN RED FONT SUGGESTS THE MODEL IS TRAINED AND TESTED ON THE SAME TARGET DOMAIN, AND WE DRAW THE RESULTS ACHIEVED BY OUR PROPOSED METHOD WITH BLUE-BOLD COLOR. THE PERFORMANCE GAINS (+)/DROPS (−) ARE LISTED IN THE BRACKET

V2V4Real → DAIR-V2X (real2real V2I)		
Methods	*V2V4Real Test Set*	
	AP@50 (%)	*AP@70 (%)*
CoBEVT *w.*/o	26.6	13.3
CoBEVT *w.*/GRL	33.5 (+6.9)	16.2 (+2.9)
CoBEVT *w.*/AdvGRL	35.8 (+9.2)	19.3 (+6.0)
CoBEVT *w.*/Ours	**36.2 (+9.6)**	**20.1 (+6.8)**
CoBEVT (*Oracle*)	60.9 (+34.3)	43.8 (+30.5)
V2X-ViT *w.*/o	31.7	12.5
V2X-ViT *w.*/GRL	34.3 (+2.6)	13.2 (+0.7)
V2X-ViT *w.*/ AdvGRL	34.1 (+2.4)	15.2 (+2.7)
V2X-ViT *w.*/Ours	**38.8 (+7.1)**	**17.2 (+4.7)**
V2X-ViT (*Oracle*)	56.1 (+24.4)	40.9 (+28.4)

D. Ablation Study

Ablation analysis is further developed to evaluate the contribution of each component. Unless explicitly specified, all experiments are conducted on **OPV2V** → **V2V4Real** setting, and we adopt *CoBEVT* [21] as the baseline for comparison.

1) Analysis on Component Contributions: To verify the effects of each component, we append three proposed methods into the vanilla CoBEVT in both V2I and V2V adaptation tasks, respectively, and quantitative results are presented in Tables V and VI. Evidently, continuous and steady accuracy improvements can be observed of each modules.

a) OPV2V → V2V4Real: CKT, BLC, and CPA modules provide 2.0%/7.1%/5.6% AP@50 increments, and similar elevation is received at 0.7 IoU threshold, indicating the effectiveness of the proposed methods. It is assumed that CKT promotes collaborative feature transferability across agents, and BLC and CPA are beneficial for location rectification and category refinement. More importantly, the integration of arbitrary designs further expedites the adaptation performance: combining CPA and BLC methods reports 40.8% AP@50 and 16.0% AP@70, and all compositions exhibit the optimal performance, reflecting their advancement and compatibility.

b) OPV2V → DAIR-V2X: In the V2I adaptation scenario, notably CKT achieves 8.9% AP@50 gains, surpassing 6.7%/6.1% of BLC and CPA. The same case occurs under the 0.7 threshold, which indicates CKT plays an even more significant role when collaborating among heterogeneous agents. Consistent with previous findings, the organic integration of our components has greatly enhanced the robustness of perception.

TABLE V

ABLATION STUDY OF *Component* CONTRIBUTIONS ON **OPV2V** → **V2V4REAL (V2V)**. NOTED THAT WE DRAW THE BEST RESULTS WITH BOLD FONT, AND PERFORMANCE GAINS (+) ARE PRESENTED IN THE BRACKET

Components			AP@50 (%)	AP@70 (%)
CKT	***BLC***	***CPA***		
			33.4	13.0
✓			35.4 (+2.0)	14.8 (+1.8)
	✓		**40.5 (+7.1)**	**15.2 (+2.2)**
		✓	39.0 (+5.6)	13.9 (+0.9)
✓	✓		39.9 (+6.5)	16.1 (+3.1)
✓		✓	40.1 (+6.7)	**16.7 (+3.7)**
	✓	✓	**40.8 (+7.4)**	16.0(+3.0)
✓	✓	✓	**44.5 (+11.1)**	**17.5 (+4.5)**

TABLE VI

ABLATION STUDY OF *Component* CONTRIBUTIONS ON **OPV2V** → **DAIR-V2X (V2I)**. NOTED THAT WE DRAW THE BEST RESULTS WITH BOLD FONT, AND PERFORMANCE GAINS (+) ARE PRESENTED IN THE BRACKET

Components			AP@50 (%)	AP@70 (%)
CKT	***BLC***	***CPA***		
			19.8	13.0
✓			**28.7 (+8.9)**	**15.9 (+2.9)**
	✓		26.5 (+6.7)	14.5 (+1.5)
		✓	25.9 (+6.1)	14.2 (+1.2)
✓	✓		32.9 (+13.1)	**16.5 (+3.5)**
✓		✓	**33.4 (+13.6)**	16.0 (+3.0)
	✓	✓	30.9 (+11.1)	15.8(+2.8)
✓	✓	✓	**35.5 (+15.7)**	**17.7 (+4.7)**

2) Analysis on Knowledge Transfer: To investigate the function of collaborative knowledge in adaptation task, we fuse different features into the same CKT for comparison. As illustrated in Table VII, aggregating two ego-vehicle f_e or joint representations $\mathcal{F}_e$ confines to a poor result. Conversely, a substantial accuracy leap is obtained by a combination of multiagent and ego features, which further improves two competitors over 1.6% and 1.7% AP growths at both IoU thresholds. It is argued that the homogeneity, e.g., f_e/f_e or $\mathcal{F}_e/\mathcal{F}_e$, may suffer from feature redundancy and noise disturbance, thus receiving an unsatisfactory result. Our fusion strategy could exploit two complementary maps to ensure the robust cross-domain knowledge transfer without key-point loss, and adequate performance gains also confirm this intuition.

3) Analysis on Bin Number: To determine the optimal of bin category, we vary the coordinate space with different intervals, and testify the performance variation as listed in Table VIII. Apparently, accuracy gain is aggressively proportional to the bin number until a 5-bin classification reveals a best adaptation of 40.5% AP@50 and 18.2% AP@70. Nonetheless, significant reduction occurs with finer partition, e.g., 15. We speculate that more detailed division might result in an implicit ambiguity or confusion during feature transfer, and five-class bin is preferably suitable for domain-adaptive collaboration.

TABLE VII

ABLATION STUDY OF *Knowledge* CONTRIBUTIONS ON **OPV2V** → **V2V4REAL**. NOTED THAT WE DRAW THE BEST RESULTS WITH BLUE-BOLD FONT, AND PERFORMANCE GAINS (+)/DROPS (−) ARE PRESENTED IN THE BRACKET

Knowledges				AP@50 (%)	AP@70 (%)
f_e	f_e	$\mathcal{F}_e$	$\mathcal{F}_e$		
				33.4	13.0
✓	✓			33.8 (+0.4)	13.1 (+0.1)
		✓	✓	33.7 (+0.3)	12.2 (−0.8)
✓			✓	**35.4 (+2.0)**	**14.8 (+1.8)**

TABLE VIII

ABLATION STUDY OF *Bin* CONTRIBUTIONS ON **OPV2V** → **V2V4REAL**. NOTED THAT WE DRAW THE BEST RESULTS WITH BLUE-BOLD FONT, AND PERFORMANCE GAINS (+)/DROPS (−) ARE PRESENTED IN THE BRACKET

Bins	AP@50 (%)	AP@70 (%)
0	33.4	13.0
2	35.9 (+2.5)	13.6 (+0.6)
3	39.6 (+6.2)	11.5 (−1.5)
5	**40.5 (+7.1)**	**18.2 (+5.2)**
10	38.8 (+5.4)	14.0 (+1.0)
15	28.7 (−4.7)	8.5 (−4.5)

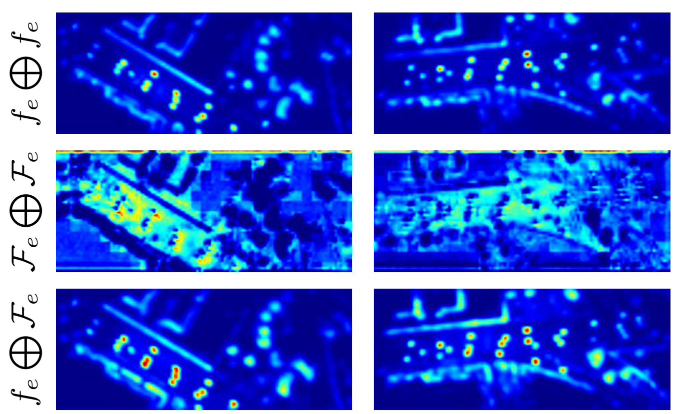

Fig. 5. Visualization of attention map on *OPV2V* → *V2V4Real* split. We list the attention map activated by the homogeneous feature fused with CoBEVT [21], i.e., $f_e \bigoplus f_e$ or $\mathcal{F}_e \bigoplus \mathcal{F}_e$, and our heterogeneous $f_e \bigoplus \mathcal{F}_e$ collaboration in CKT module. Noted that the brighter spots highlight the more significant/potential target area. Best viewed in color.

E. More Visualizations

In this section, visualization results are presented to explore how each ingredient of ***CUDA-X*** makes effect on feature and candidate adaptation across datasets straightforwardly.

1) Attention Maps in Feature Adaptation: Considering the prevalent CoBEVT [21] in OPV2V → V2V4Real setting, we depict the attention maps activated by different knowledge collaborations, i.e., $f_e \bigoplus f_e$, $\mathcal{F}_e \bigoplus \mathcal{F}_e$, and $f_e \bigoplus \mathcal{F}_e$.

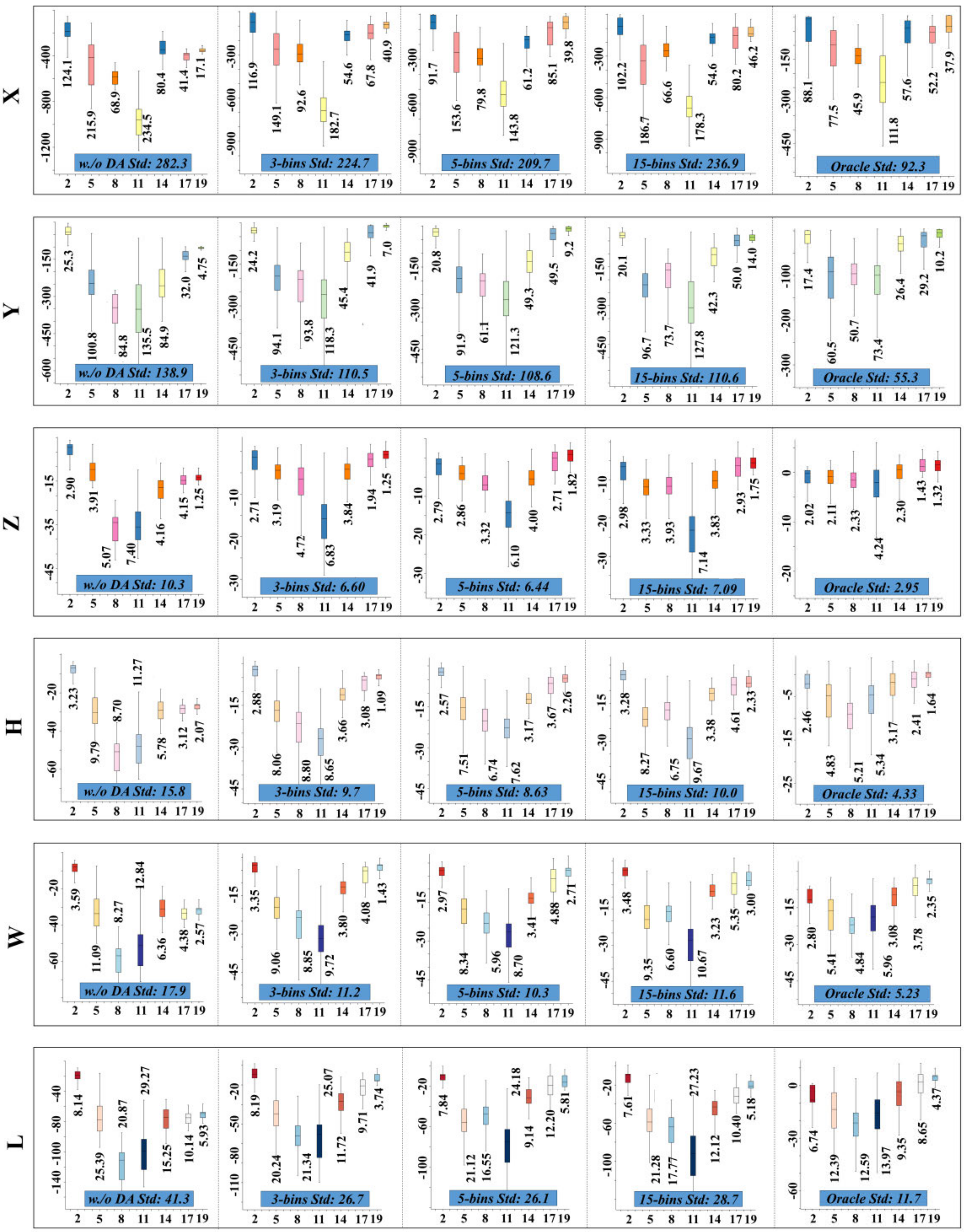

Fig. 6. Visualization of residual boxplots on *OPV2V* → *V2V4Real* (sim2real) test split. We investigate how various bin categories with CoBEVT [21] contribute to std fluctuation during box regression, and list the residuals along 6 spatial coordinates from the left to right column. Best viewed in color.

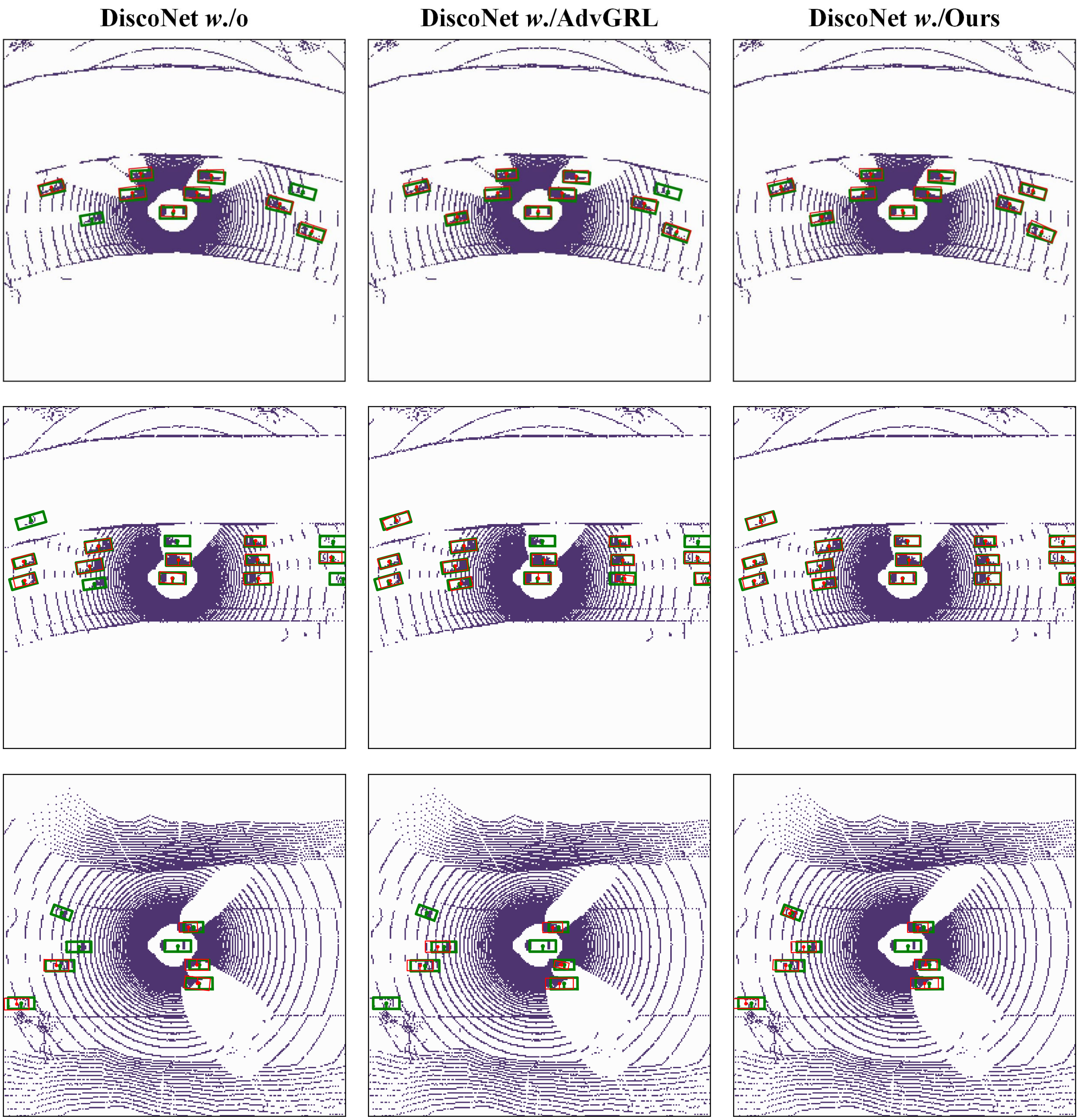

Fig. 7. Visualization of detection results on *V2X-Sim* → *OPV2V* (*sim2sim V2V*) test split. We list the detection performance achieved by DiscoNet [19] with no-adaptation, AdvGRL, and our CUDA-X from the left to right column across various scenes, respectively. Noted that a pair of GT box and its prediction is drawn with *green* and *red* rectangles. Best viewed in color.

As presented in Fig. 5, it is noted that homogeneous feature aggregation introduces unavoidable background interference and sensor noise, particularly for $\mathcal{F}_e \bigoplus \mathcal{F}_e$. Comparatively, our $f_e \bigoplus \mathcal{F}_e$ emphasizes the potential vehicle location with apparent foreground/background contour, and more brighter the color is, more higher probability the target occurs. We confirm that spatial–channel operation not only acts as an information filter for irrelevance removal, but it is essentially endowed with the collaborative knowledge transfer. Moreover, key-point combination is beneficial to domain-agnostic feature extraction, thus being more robust against the cross-domain divergence.

2) Residual Boxplot in Candidate Adaptation: The number of bin category in CoBEVT is meticulously inspected during OPV2V → V2V4Real circumstance, and boxplots of residual between the predicted and GT rectangles are illustrated in Fig. 6 with respect to six coordinates. Distinctly, 5-bin classification reports a solid yet bearable Std fluctuation than other

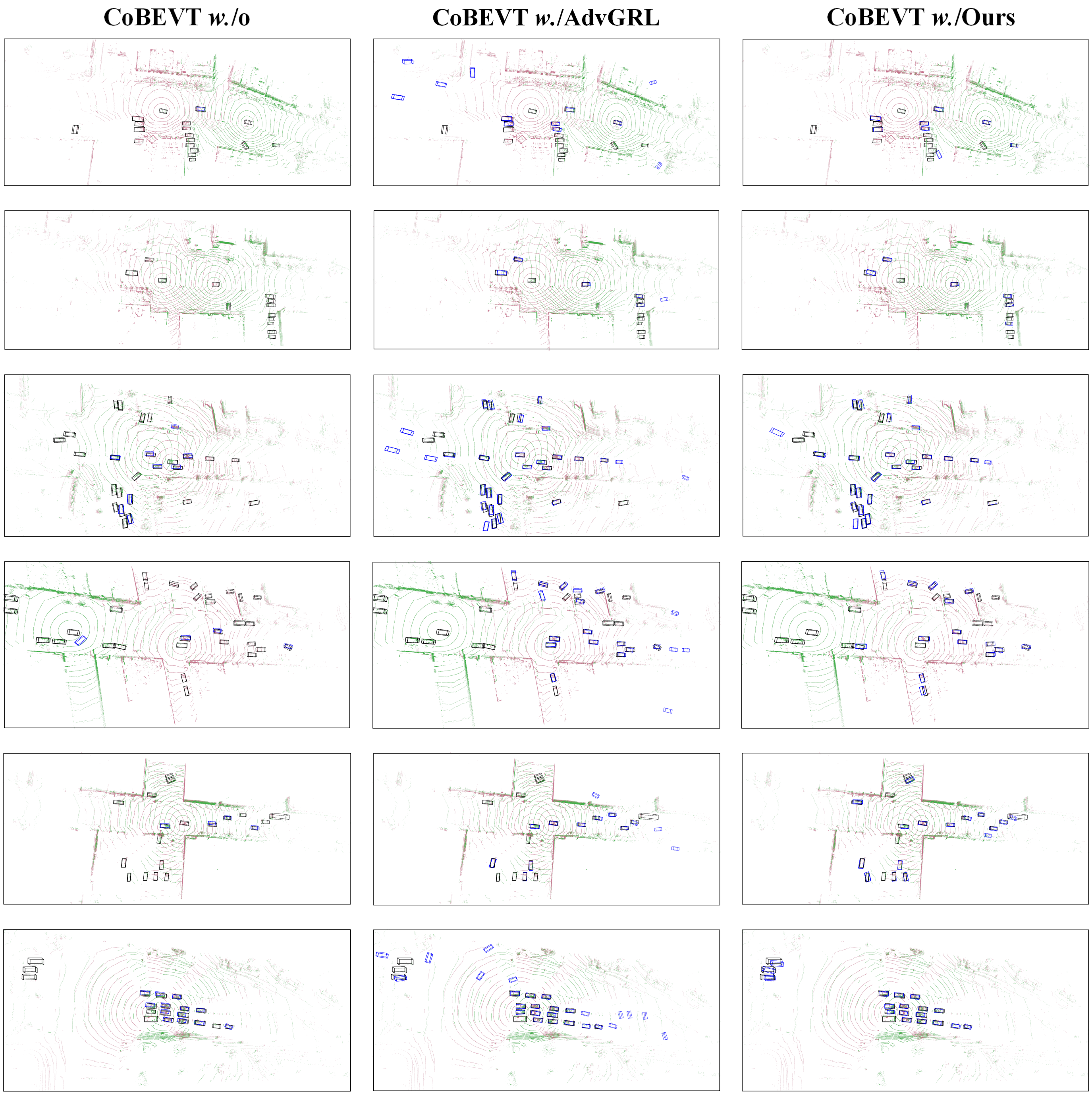

Fig. 8. Visualization of detection results on *OPV2V* → *V2V4Real* (*sim2real V2V*) test split. We list the detection performance achieved by CoBEVT [21] with no-adaptation, AdvGRL, and our CUDA-X from the left to right column, under a diversity of scenarios for each row, e.g., straight, curve, intersection, and so on. Noted that we render the ego-centric LiDAR data with *red* color, and nearby CAVs with *green* color for clarification, and a pair of GT box and its prediction is drawn with *black* and *blue* rectangles. Best viewed in color.

competitors in all spatial dimensions, which is consistent with the above-mentioned clarification in Table VII. It is argued that excessive rough or fine-grained division is detrimental to box regression, while 5-class bin with reasonable interval is more appropriate for our adaptation setting.

3) Detection Results: We ultimately outline 3-D object detection visualizations achieved by DiscoNet [19], CoBEVT [21], and V2V-ViT [40] with various adaptations among V2X-Sim → OPV2V (Fig. 7), OPV2V → V2V4Real (Fig. 8), and V2V4Real → DAIR-V2X (Fig. 9) test splits. Intuitively, CUDA-X evidently demonstrates unprecedented and extraordinary performance compared to the counterparts. With the introduction of the proposed method, DiscoNet could localize the driving vehicles precisely in the OPV2V curve scene; meanwhile, CoBEVT receives higher precision–recall and less false-positive predictions with regards to a diversity of scenarios in V2V4Real samples; V2X-ViT $w.$/Ours realizes the leading cross-agent result in the actual environment, uniformly verifying its advancement and superiority.

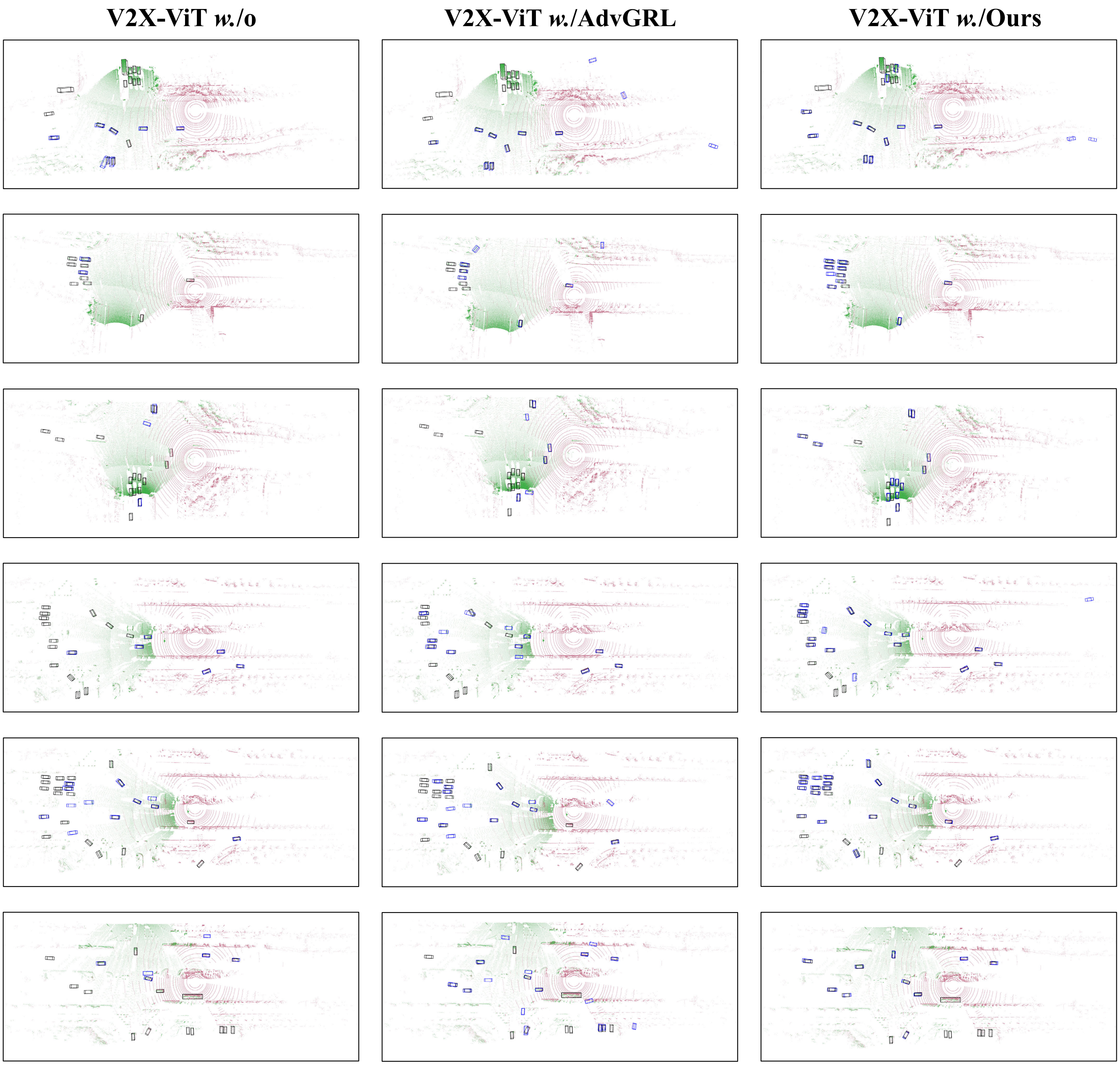

Fig. 9. Visualization of detection results on *V2V4Real* → *DAIR-V2X* (*real2real V2I*) test split. We list the detection performance achieved by V2X-ViT [16] without adaptation, AdvGRL, and our ***CUDA-X*** from the left to right column across various scenes, respectively. Noted that we render the ego-centric LiDAR data with *red* color, and nearby infrastructure with *green* color for clarification, and a pair of GT box and its prediction is drawn with *black* and *blue* rectangles. Best viewed in color.

V. Conclusion

In this article, we develop CUDA-X perception that is constructed on the top of an off-the-shelf cooperative perception model with multiagent feature distillation and instance candidate alignment. To be specific, CKT is responsible for domain-agnostic feature reconstruction from nearby agents, and BLC with CPA enforce an auxiliary supervision for cross-domain box coordination refinement and instance-level category-specific adaptation. Empirical study demonstrates the state-of-the-art performance on **V2V** (V2X-Sim → OPV2V and OPV2V → V2V4Real) and **V2I** (OPV2V → DAIR-V2X and V2V4Real → DAIR-V2X) adaptations. We hope our work could shed light on exploring more generalizable multiagent perception, and more investigation of extreme variation in data distribution or long-tailed scenario would be conducted in the future.

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